
Security Monitoring of Active Directory Infrastructure using Wazuh SIEM

Date: March 2026

1. Executive Summary

This report documents the design, deployment, and ongoing monitoring of an Active Directory (AD) infrastructure using Wazuh SIEM within a purpose-built home lab environment. The goal of this project is to gain hands-on experience with enterprise-grade identity infrastructure and to understand how a modern, open-source SIEM platform monitors and protects it.

The lab consists of a Windows Server 2025 Domain Controller, a Windows 10 Enterprise domain-joined client machine, and a Wazuh Manager running on Ubuntu 22.04 via WSL2. Wazuh agents are deployed on both Windows machines and are actively collecting logs, security events, and file integrity data in real time.

Scope Note

This report covers infrastructure setup, Wazuh integration, and passive security monitoring only. No attack simulations or adversarial techniques have been executed in this environment at this stage. The focus is on establishing a solid, observable baseline and understanding what normal AD activity looks like through the lens of Wazuh.

The project demonstrates that even a minimal two-machine AD lab, when properly instrumented with Wazuh, produces rich security telemetry covering authentication events, group policy changes, privilege usage, and directory service access — providing a realistic foundation for future security use-case development.

2. Problem Statement

2.1 Background

Active Directory is the cornerstone of identity and access management in Windows environments, deployed across most enterprise networks worldwide. Despite its ubiquity, AD is one of the most poorly monitored components in many organizations — primarily because the volume and complexity of its event logs make manual analysis impractical without dedicated tooling.

A SIEM solution bridges this gap by aggregating, normalizing, and correlating log data from across the environment, surfacing anomalies and policy violations that would otherwise go unnoticed. Wazuh, as a free and open-source SIEM and XDR platform, represents a practical and cost-effective choice for this purpose — particularly in research and small-enterprise contexts.

2.2 Core Challenges Addressed

- Lack of visibility into authentication events, logon failures, and privilege escalation on Windows endpoints
- No centralized aggregation of Windows Security Event Logs without a dedicated log management platform
- Difficulty establishing a behavioural baseline for normal AD operations without monitoring tooling
- Understanding which Wazuh out-of-the-box rules fire in a clean, attack-free AD environment
- Learning the end-to-end pipeline from Windows event generation to SIEM alert in a controlled setting

2.3 Project Objectives

- Deploy a functional Active Directory domain using Windows Server 2025
- Join a Windows 10 Enterprise client to the domain and configure it as a managed endpoint
- Install and configure Wazuh Manager with agents on both Windows machines
- Enable comprehensive audit policies to maximize event log visibility
- Understand the types of alerts generated during normal AD operations
- Document the architecture, deployment process, and monitoring outputs as a reference for future work

3. Lab Environment Design

3.1 Overview

The lab is a minimal but realistic representation of a small enterprise Windows environment. It uses VMware Workstation as the hypervisor to host all machines on a custom virtual network, with the Wazuh Manager running inside a WSL2 (Windows Subsystem for Linux) Ubuntu instance on the host machine. This approach reduces resource consumption while maintaining full network connectivity between components.

3.2 Hardware Specifications

Host Machine	Personal workstation running Windows 11 with VMware Workstation Pro
Hypervisor	VMware Workstation Pro
Network Mode	VMware Custom Network (192.168.114.0/24) for VM-to-VM communication
Wazuh Host	WSL2 Ubuntu 22.04 LTS running directly on the physical host machine

3.3 Virtual Machine Inventory

VM Name	OS / Role	IP Address
Windows Server 2025	Windows Server 2025 — Primary Domain Controller (DC)	192.168.114.130
Windows 10	Windows 10 Enterprise — Domain-Joined Client Workstation	192.168.114.129
WAZUH-WSL	Ubuntu 22.04 LTS (WSL2) — Wazuh Manager 4.7	172.25.97.95

3.4 Domain Configuration

Domain Name	Nexus.corp
Forest / Domain Level	Windows Server 2016 Functional Level
Domain Controller	Windows Server 2025 (single DC)
Organizational Units	IT, HR, Sales
Group Policy Objects	Default Domain Policy, Password Policy, Account Lockout Policy, Audit Policy, Drive Mapping
DNS	Integrated DNS running on the Domain Controller
Client Machine	Windows 10 Enterprise joined to Nexus.corp

3.5 User Accounts Created

Account	Role / Description
Administrator	Built-in domain administrator account
Sales, Varun Shah	Standard domain users — Sales OU
HR, Ajay Singh	Standard domain user — HR OU
IT	Standard domain user — IT OU

4. Architecture & Network Diagram

4.1 High-Level Architecture

The environment uses a hub-and-spoke monitoring model. The Wazuh Manager, running in WSL2, acts as the central collection and analysis engine. Wazuh agents installed on the Domain Controller and the Windows 10 client forward security events, log data, and file integrity information to the manager over an encrypted TCP channel.

Network Topology

The Domain Controller (192.168.114.130) and the Windows 10 client (192.168.114.129) are connected on a VMware custom virtual network (192.168.114.0/24). The Wazuh Manager in WSL2 (172.25.97.95) is reachable from both VMs via the WSL2 virtual adapter. The Wazuh Dashboard is accessible from the host browser at <https://172.25.97.95>.

4.2 Component Communication Flow

Source	Destination	Protocol / Port
Wazuh Agent (DC — WS2025)	Wazuh Manager (WSL2)	TCP 1514 (encrypted)
Wazuh Agent (Win10 Client)	Wazuh Manager (WSL2)	TCP 1514 (encrypted)
Wazuh Manager	Wazuh Indexer	TCP 9200 (REST API)
Wazuh Indexer	Wazuh Dashboard	TCP 9200
Host Browser	Wazuh Dashboard	HTTPS 443

4.3 Data Flow

- The Wazuh agent on the Domain Controller collects Windows Security, System, Application, and Directory Service event logs
 - The Wazuh agent on the Windows 10 client collects Security, System, Application, and PowerShell Operational logs
 - Log data is forwarded over encrypted TCP (port 1514) to the Wazuh Manager in WSL2
 - The Wazuh Manager decodes events, matches them against rule sets, and generates alerts
 - Alerts and raw events are stored in the Wazuh Indexer (OpenSearch) for querying and retention
 - The Wazuh Dashboard (accessible via browser) provides real-time alert views, agent status, and compliance dashboards
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Active Directory Logical Structure

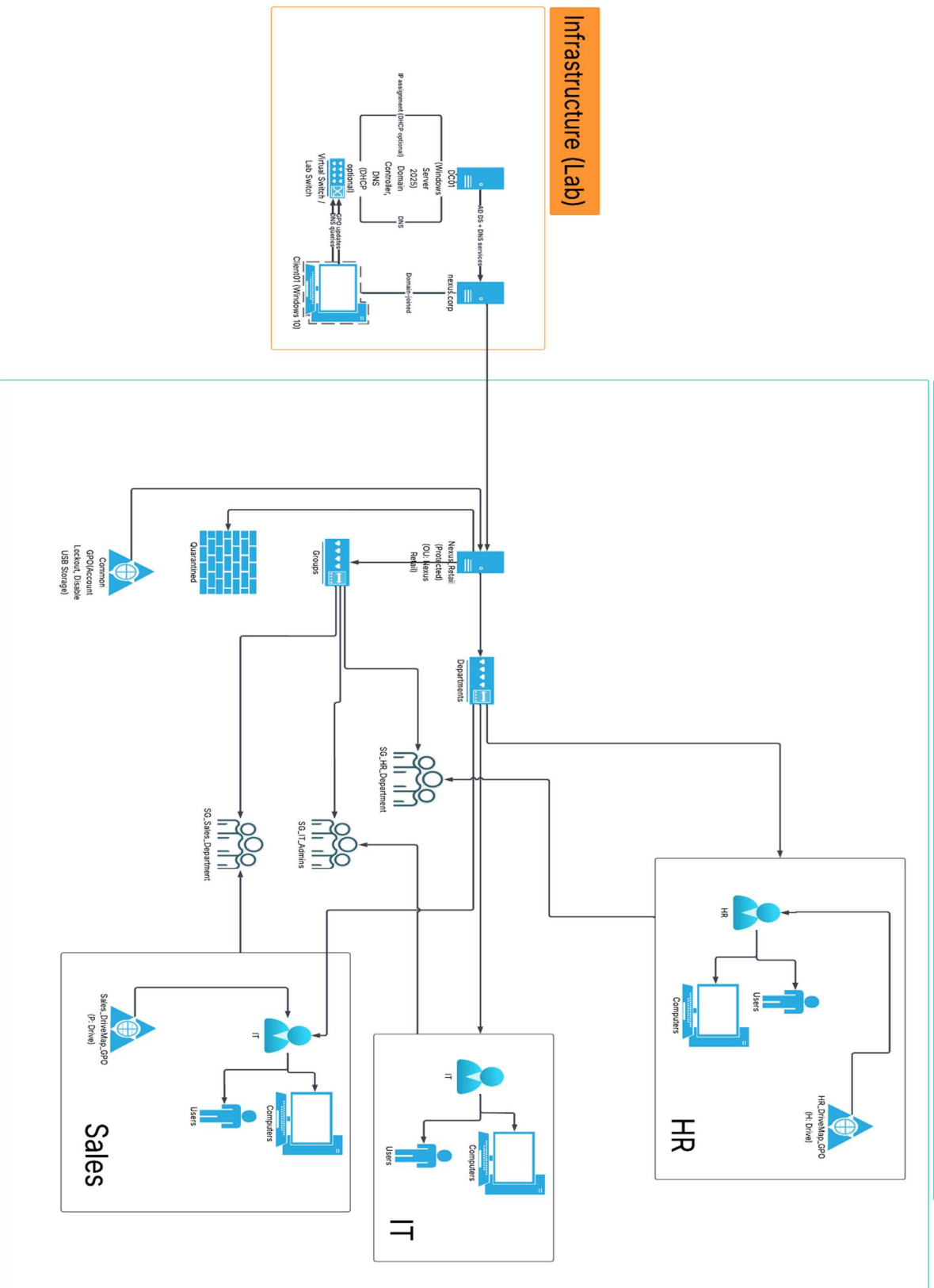


Figure 4.1

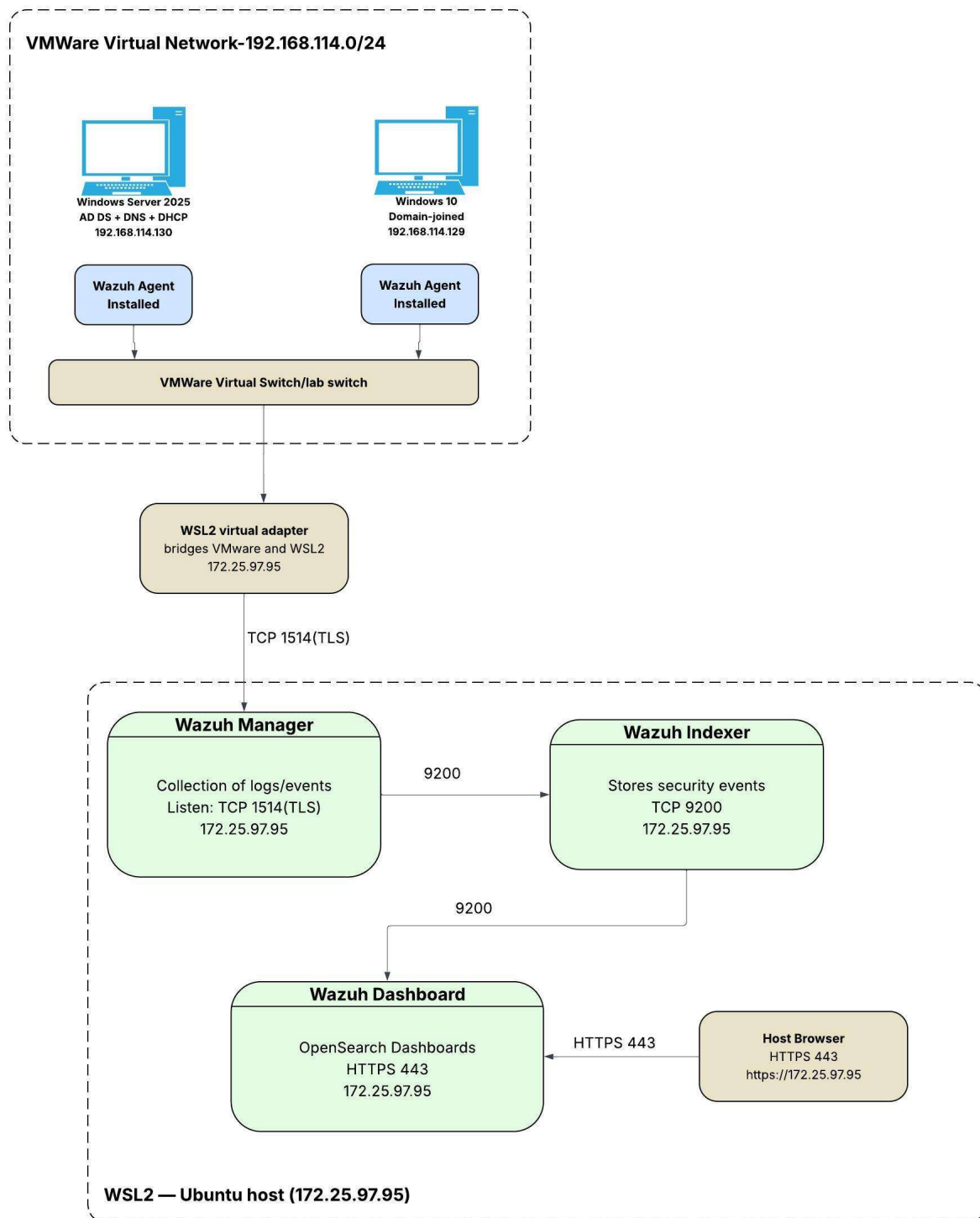


Figure 4.2

5. Deployment Process

5.1 Active Directory Setup — Windows Server 2025

Windows Server 2025 was installed on a VMware VM and promoted to a Domain Controller for the Nexus.corp domain. The following roles and features were configured:

- Active Directory Domain Services (AD DS) —installed and promoted to DC.
- DNS Server — integrated with AD for name resolution
- Group Policy Management Console (GPMC) — for policy authoring and linking
- Active Directory Users and Computers (ADUC) — for OU and user management

Server Name	WIN-DC01
OS Version	Windows Server 2025 Datacenter Evaluation
IP Address	192.168.114.130 (static)
Domain Role	Primary Domain Controller
Domain	Nexus.corp
DNS	Self-hosted (127.0.0.1 as preferred DNS)

5.2 Windows 10 Client — Domain Join

A Windows 10 Enterprise VM was deployed and joined to the Nexus.corp domain. DNS was pointed to the Domain Controller (192.168.114.130) to enable successful domain resolution and join.

Machine Name	WIN-CLIENT01
OS Version	Windows 10 Enterprise (22H2)
IP Address	192.168.114.129 (static)
Domain	Nexus.corp
Primary DNS	192.168.114.130 (Domain Controller)
Logged-on User	Nexus\HR (standard domain user)

5.3 Wazuh Manager Installation — WSL2 Ubuntu

Wazuh was deployed using the official all-in-one installer script on Ubuntu 22.04 LTS running inside WSL2. The installer sets up the Wazuh Manager, Wazuh Indexer (OpenSearch), and Wazuh Dashboard as an integrated stack.

Installation Commands

```
# Step 1: Download the Wazuh all-in-one installer
curl -sO https://packages.wazuh.com/4.7/wazuh-install.sh

# Step 2: Run the installer (deploys Manager + Indexer + Dashboard)
sudo bash ./wazuh-install.sh -a

# Step 3: Verify all three services are active
sudo systemctl status wazuh-manager wazuh-indexer wazuh-dashboard

# Retrieve the auto-generated admin credentials
sudo tar -O -xvf wazuh-install-files.tar wazuh-install-files/wazuh-passwords.txt
```

Wazuh Manager IP	172.25.97.95 (WSL2 virtual adapter)
Dashboard URL	https://172.25.97.95 (accessible from host browser)
Agent Port	TCP 1514 (agent communication)
API Port	TCP 55000 (Wazuh REST API)
Indexer Port	TCP 9200 (OpenSearch)

5.4 Wazuh Agent Deployment — Windows Machines

Wazuh agents were installed manually on both Windows machines using the MSI installer downloaded from the Wazuh documentation. Each agent was configured to point to the Wazuh Manager IP address and registered using the agent enrolment process.

Agent Installation (PowerShell — run as Administrator)

```
# Download Wazuh Agent MSI for Windows x64
Invoke-WebRequest -Uri https://packages.wazuh.com/4.x/windows/wazuh-agent-4.7.0-1.msi `
  -OutFile 'C:\wazuh-agent.msi'

# Install silently, pointing to the Wazuh Manager
msiexec.exe /i C:\wazuh-agent.msi /q `
  WAZUH_MANAGER='172.25.97.95' `
  WAZUH_AGENT_GROUP='windows' `
  WAZUH_REGISTRATION_SERVER='172.25.97.95'

# Start the Wazuh agent service
NET START Wazuh
```

Agent on DC	WIN-DC01 — registered, active, forwarding logs
Agent on Client	WIN-CLIENT01 — registered, active, forwarding logs
Agent Version	Wazuh Agent 4.14.x for Windows (x64)

Log Sources (DC)	Security, System, Application, Directory Service, DNS Server
Log Sources (Client)	Security, System, Application, PowerShell Operational

5.5 Audit Policy Configuration

Advanced Audit Policy was enabled via Group Policy on the Domain Controller and applied to all domain-joined machines. This ensures that Windows generates the detailed event log entries that Wazuh relies on for detection and monitoring.

Audit Category	Policy Setting
Account Logon — Credential Validation	Success and Failure
Account Logon — Kerberos Authentication	Success and Failure
Account Management — User Account Management	Success and Failure
Account Management — Security Group Management	Success and Failure
DS Access — Directory Service Changes	Success and Failure
DS Access — Directory Service Access	Success and Failure
Logon/Logoff — Logon	Success and Failure
Logon/Logoff — Special Logon	Success and Failure
Object Access — SAM	Success and Failure
Privilege Use — Sensitive Privilege Use	Success and Failure
Policy Change — Audit Policy Change	Success and Failure
System — Security System Extension	Success and Failure

6. Security Use-Cases Implemented

The following monitoring use-cases are active in the current lab configuration. These are based on Wazuh out-of-the-box rules combined with the audit policy settings configured in Section 5.5. All use-cases reflect passive monitoring of legitimate AD activity — no attack simulation has been performed.

6.1 Authentication & Logon Monitoring

- Successful and failed interactive logons on the Domain Controller and Windows 10 client (Event IDs 4624, 4625)
- Network logon events from domain-joined machines to the DC (Kerberos / NTLM)
- Special privilege logon detection — alerting when accounts with administrative privileges log on (Event ID 4672)
- Account lockout events and lockout source tracking (Event ID 4740)
- Explicit credential usage detection (Event ID 4648) — e.g. Run As scenarios

6.2 Active Directory Object Change Monitoring

- User account creation, modification, and deletion (Event IDs 4720, 4722, 4723, 4725, 4726)
- Security group membership changes — including additions to privileged groups (Event IDs 4728, 4732, 4756)
- Password reset events, both admin-initiated and user self-service (Event ID 4723, 4724)
- OU and directory object modifications tracked via Directory Service Change audit (Event ID 5136)
- Computer account creation and deletion — e.g. when new machines join the domain (Event ID 4741, 4743)

6.3 Group Policy Monitoring

- GPO creation and modification events captured via Directory Service Changes audit (Event IDs 5136, 5137, 5141)
- Group Policy application success and failure events on client machines (Event IDs 1500–1503 from Group Policy log)
- Wazuh alerts on unexpected changes to security-relevant GPOs such as password policy or audit policy objects

6.4 Privilege and Service Account Monitoring

- Service account logon events monitored to detect interactive logins for service-type accounts (should be non-interactive)
- Privilege use events — sensitive privileges such as SeDebugPrivilege, SeTcbPrivilege tracked via Event ID 4673
- Scheduled task creation by non-system accounts monitored on both DC and client (Event ID 4698)

6.5 System Integrity and Configuration Monitoring

- File Integrity Monitoring (FIM) enabled on critical system directories: C:\Windows\System32, C:\Windows\SYSVOL, and NTDS database path
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- Wazuh FIM detects any new, modified, or deleted files in monitored paths and generates alerts accordingly
 - Windows Firewall and service start/stop events monitored for baseline configuration changes (Event IDs 7035, 7036, 7040)
 - Registry monitoring for critical keys under HKLM\SYSTEM and HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion

6.6 Wazuh Built-in Compliance Dashboards

- PCI DSS compliance view — highlights events related to access control, audit logging, and account management
- MITRE ATT&CK framework view — maps observed events to relevant ATT&CK techniques even during normal operations
- CIS Benchmark SCA (Security Configuration Assessment) scan run against both Windows machines to identify misconfigurations

7. Detection & Monitoring Results

7.1 Baseline Monitoring Observations

Since no attacks have been simulated, the alerts and events generated represent the normal, expected operational activity of an Active Directory environment. This baseline period is valuable — it establishes what typical AD traffic looks like and reveals which default Wazuh rules fire regularly in a healthy environment.

Baseline Value

Understanding the alert profile of a clean environment is a prerequisite for effective threat detection. Events observed during this phase establish the noise floor — the known-good patterns against which future anomalies will be measured.

7.2 Alert Distribution — Baseline Period

The following table reflects representative alert counts observed during the initial monitoring period across both monitored agents (DC + Win10 client):

Severity Level	Typical Sources	Example Rule Triggers
High	Authentication events, privilege use	Multiple failed logins, special privilege login
Medium	Account management, GPO, service events	User account modified, service started/stopped

Low	Normal logon activity, FIM baseline writes	Successful logon, file accessed in monitored path
Info	Agent heartbeats, policy refresh, DNS	Group Policy applied, agent connected, DNS query

7.3 Events Observed on Domain Controller (WIN-DC01)

- Frequent Kerberos TGT and TGS ticket requests (Event ID 4768, 4769) — normal domain authentication activity
- NTLM authentication events where older protocols fall back (Event ID 4776)
- Periodic Group Policy refresh events from the Windows 10 client (every ~90 minutes by default)
- Directory Service access events when LDAP queries are issued by the client machine
- Administrator logon and logoff events during DC management tasks
- DNS server events — query resolution for Nexus.corp and external lookups

7.4 Events Observed on Windows 10 Client (WIN-CLIENT01)

- Interactive domain user logon (HR) — Event ID 4624, Logon Type 2
- Network logon events when accessing shared resources on the DC
- PowerShell script block logging events during any PS activity in the session
- Windows Defender real-time protection activity events
- Scheduled Task activity — built-in Windows maintenance tasks
- User-initiated password change event (Event ID 4723) during lab testing

7.5 Wazuh SCA Results

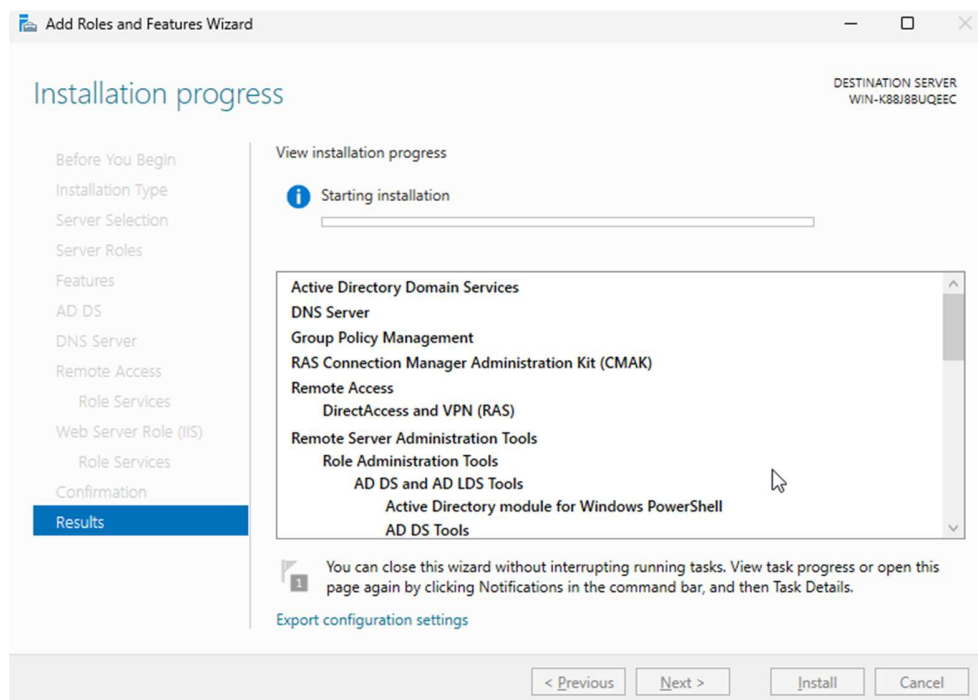
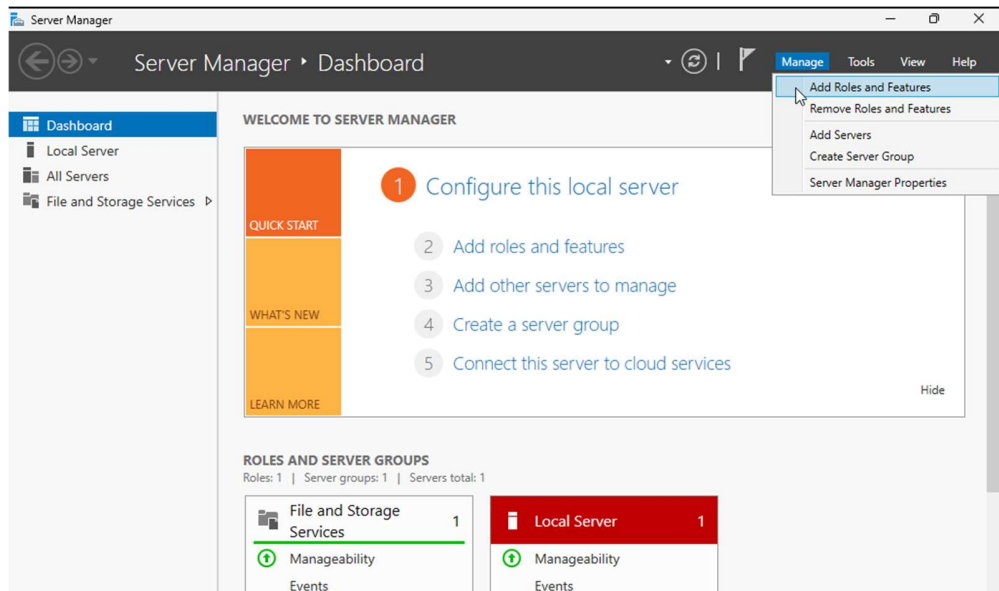
Wazuh's Security Configuration Assessment (SCA) module ran an automated benchmark check against both machines using the CIS Windows benchmark policy. Key findings from the initial scan:

SCA Check Category	Result Summary
Password Policy	Passed — minimum length and complexity enforced via GPO
Account Lockout Policy	Passed — lockout threshold set via Default Domain Policy
Audit Policy	Passed — advanced audit policy enabled on DC and propagated
Guest Account Status	Passed — Guest account disabled on both machines
Windows Update	Info — patch level should be reviewed for both VMs

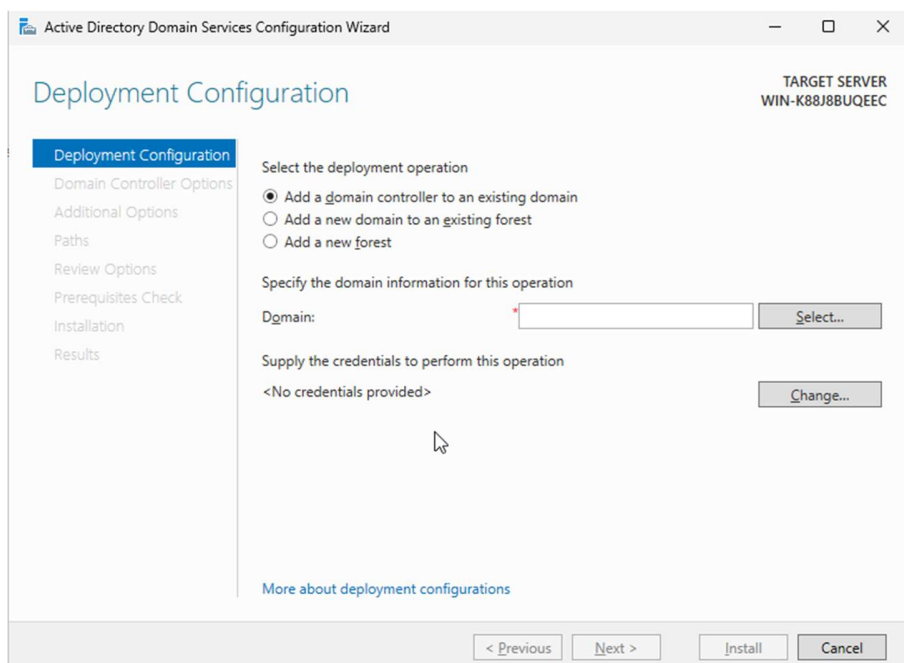
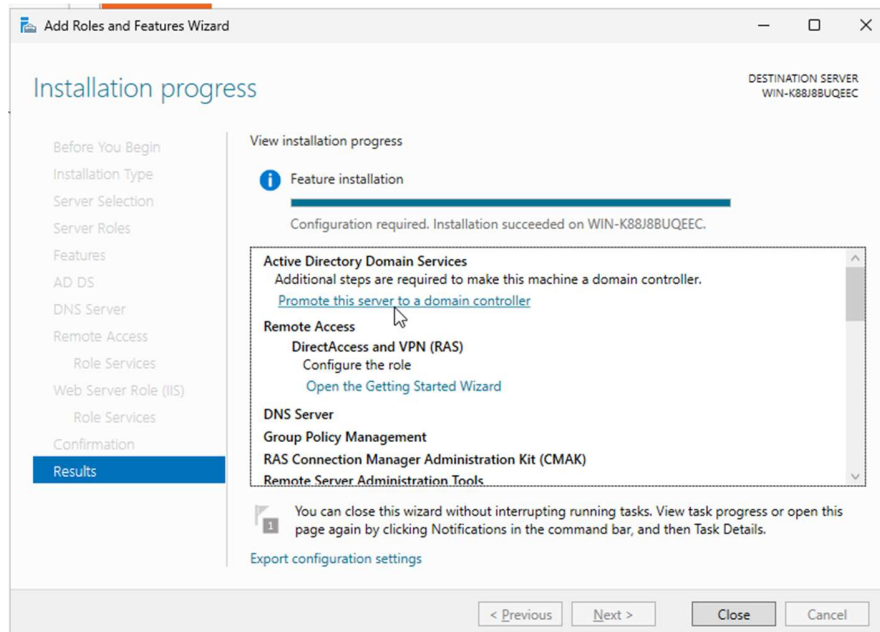
Title	Target	Result
Ensure 'Relax minimum password length limits' is set to 'Enabled'.	Registry: HKEY_LOCAL_MACHINE\System\CurrentControlSet\Control\SAM	● Passed
Ensure 'Account lockout threshold' is set to '5 or fewer invalid logon attempt(s),...	Command: powershell secedit /export /cfg \$env:TEMP\secpol.cfg; Get-Content \$env:TEMP\secpol.cfg Select-String "LockoutBadCount"; Remove-Item \$env:TEMP\secpol.cfg	● Passed
Ensure 'Reset account lockout counter after' is set to '15 or more minute(s)'.	Command: powershell secedit /export /cfg \$env:TEMP\secpol.cfg; Get-Content \$env:TEMP\secpol.cfg Select-String "ResetLockoutCount"; Remove-Item \$env:TEMP\secpol.cfg	● Passed
Ensure 'Accounts: Limit local account use of blank passwords to console logon ...	Registry: HKEY_LOCAL_MACHINE\System\CurrentControlSet\Control\Lsa	● Passed
Configure 'Accounts: Rename administrator account'.	Command: net user	● Passed
Ensure 'Audit: Force audit policy subcategory settings (Windows Vista or later) ...	Registry: HKEY_LOCAL_MACHINE\System\CurrentControlSet\Control\Lsa	● Passed
Ensure 'Audit: Shut down system immediately if unable to log security audits' is ...	Registry: HKEY_LOCAL_MACHINE\System\CurrentControlSet\Control\Lsa	● Passed
Ensure 'Devices: Prevent users from installing printer drivers' is set to 'Enabled'.	Registry: HKEY_LOCAL_MACHINE\System\CurrentControlSet\Control\Print\Providers\Lan Man Print Services\Servers	● Passed
Ensure 'Domain controller: Allow server operators to schedule tasks' is set to 'D...	Registry: HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\Lsa	● Passed
Ensure 'Domain controller: Allow vulnerable Netlogon secure channel connectio...	Registry: HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services\Netlogon\Parame ters	● Passed
Ensure 'Domain controller: LDAP server channel binding token requirements' is ...	Registry: HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services\NTDS\Parameters	● Passed
Ensure 'Domain controller: LDAP server signing requirements Enforcement' is s...	Registry: HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services\NTDS\Parameters	● Passed
Ensure 'Domain controller: Refuse machine account password changes' is set t...	Registry: HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services\Netlogon\Parame ters	● Passed

8. Snapshots

8.1 Installation of Active Directory on Windows Server 2025



8.2 Promoting the Server to the Domain Controller (DC)



Active Directory Domain Services Configuration Wizard

TARGET SERVER
WIN-K88J8BUQEEC

Additional Options

Deployment Configuration
Domain Controller Options
DNS Options
Additional Options
Paths
Review Options
Prerequisites Check
Installation
Results

Verify the NetBIOS name assigned to the domain and change it if necessary

The NetBIOS domain name:

[More about additional options](#)

< Previous Next > Install Cancel

Active Directory Domain Services Configuration Wizard

TARGET SERVER
WIN-K88J8BUQEEC

Domain Controller Options

Deployment Configuration
Domain Controller Options
DNS Options
Additional Options
Paths
Review Options
Prerequisites Check
Installation
Results

Select functional level of the new forest and root domain

Forest functional level:

Domain functional level:

Specify domain controller capabilities

☒ Domain Name System (DNS) server
☒ Global Catalog (GC)
☐ Read only domain controller (RODC)

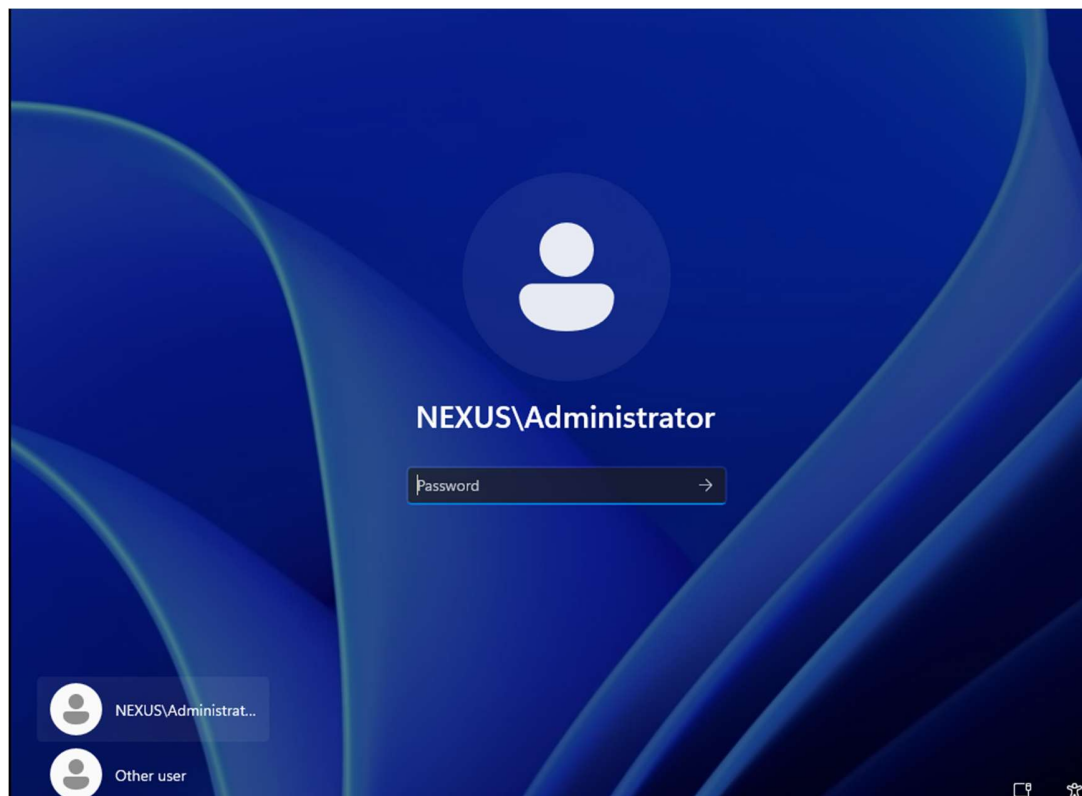
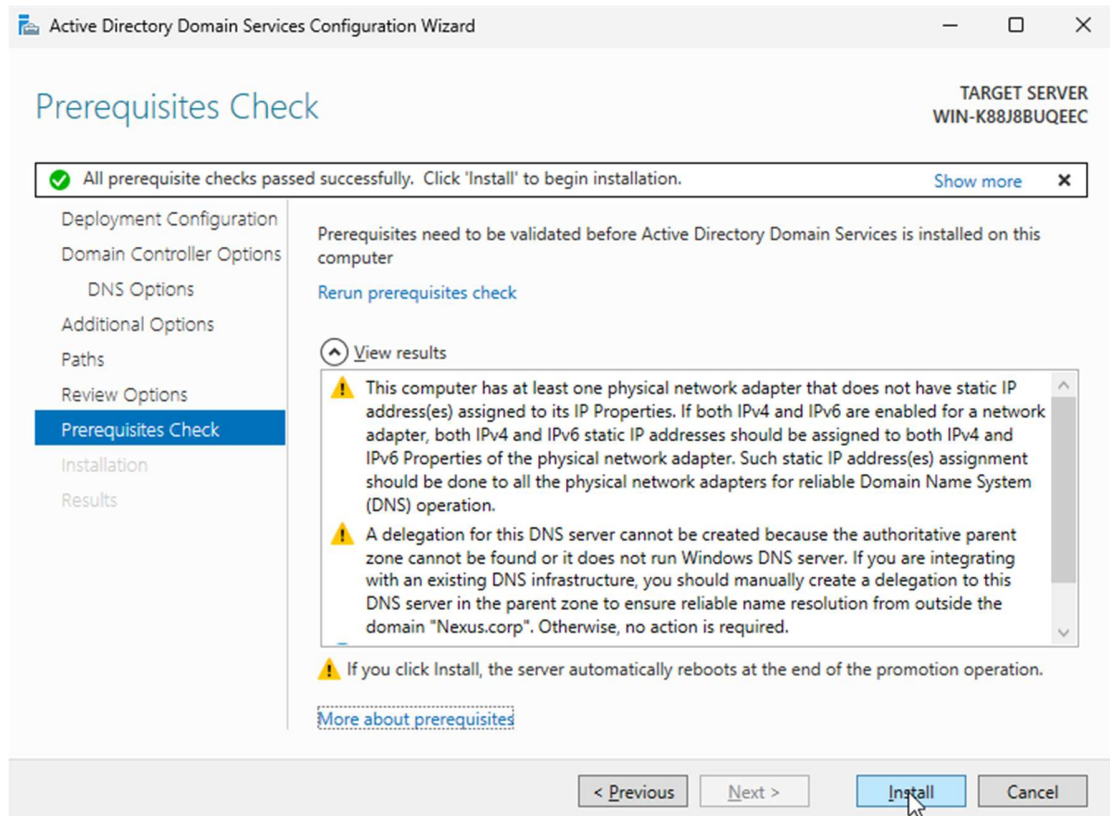
Type the Directory Services Restore Mode (DSRM) password

Password:

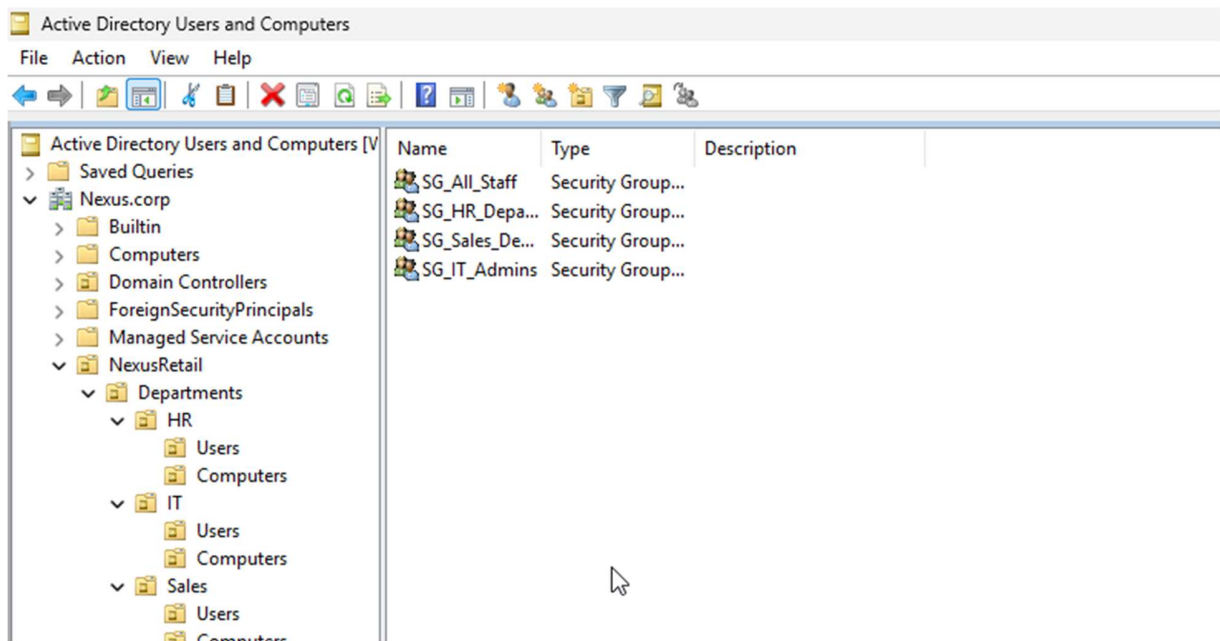
Confirm password:

[More about domain controller options](#)

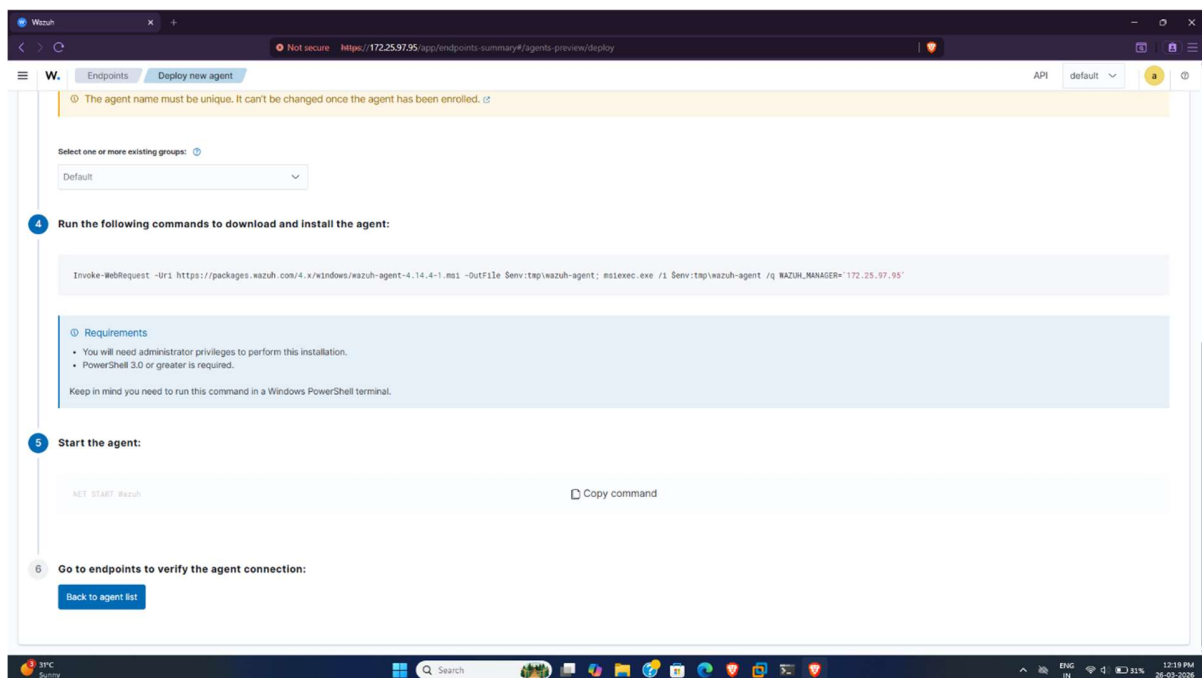
< Previous Next > Install Cancel



8.3 Active Directory Structure



8.4 Wazuh Agent Installation



Services (Local)

Wazuh

[Stop the service](#)
[Restart the service](#)

Description:
Wazuh Windows Agent

Name	Description	Status	Startup Type	Log On As
Update Orchestrator Service	Manages W...	Running	Automatic (D...	Local Syste...
UPnP Device Host	Allows UPn...	Disabled	Automatic (D...	Local Service
User Access Logging Service	This service ...	Running	Automatic (D...	Local Syste...
User Data Access_132b58	Provides ap...	Manual	Manual	Local Syste...
User Data Storage_132b58	Handles sto...	Manual	Manual	Local Syste...
User Experience Virtualizatio...	Provides su...	Disabled	Automatic (D...	Local Syste...
User Manager	User Manag...	Running	Automatic (T...	Local Syste...
User Profile Service	This service ...	Running	Automatic	Local Syste...
Virtual Disk	Provides m...	Running	Manual	Local Syste...
Volume Shadow Copy	Manages an...	Manual	Manual	Local Syste...
W3C Logging Service	Provides W...	Manual	Manual	Local Syste...
WaaSMedicSvc	<Failed to R...	Manual	Manual	Local Syste...
WalletService	Hosts objec...	Manual	Manual	Local Syste...
Warp JIT Service	Enables JIT ...	Manual (Trig...	Manual (Trig...	Local Service
Wazuh	Wazuh Win...	Running	Automatic	Local Syste...
Web Account Manager	This service ...	Running	Manual	Local Syste...
Wi-Fi Direct Services Conne...	Manages co...	Manual (Trig...	Manual (Trig...	Local Service
Windows Audio	Manages au...	Running	Automatic	Local Service
Windows Audio Endpoint B...	Manages au...	Running	Automatic	Local Syste...
Windows Biometric Service	The Windo...	Manual (Trig...	Manual (Trig...	Local Syste...
Windows Camera Frame Se...	Enables mul...	Manual (Trig...	Manual (Trig...	Local Service
Windows Camera Frame Se...	Monitors th...	Manual (Trig...	Manual (Trig...	Local Syste...
Windows Connect Now - C...	WCNCSCV ...	Manual	Manual	Local Service
Windows Connection Mana...	Makes auto...	Running	Automatic (T...	Local Service
Windows Defender Advanc...	Windows D...	Manual	Manual	Local Syste...
Windows Defender Firewall	Windows D...	Running	Automatic	Local Service
Windows Encryption Provid...	Windows E...	Manual (Trig...	Manual (Trig...	Local Service
Windows Error Reporting Se...	Allows error...	Manual (Trig...	Manual (Trig...	Local Syste...
Windows Event Collector	This service ...	Manual	Manual	Network S...

8.5 Wazuh SIEM

W. Endpoints API default 9

AGENTS BY STATUS

- Active (2)
- Disconnected (0)
- Pending (0)
- Never connected (0)

TOP 5 OS

- windows (2)

TOP 5 GROUPS

- ActiveDirectory (2)

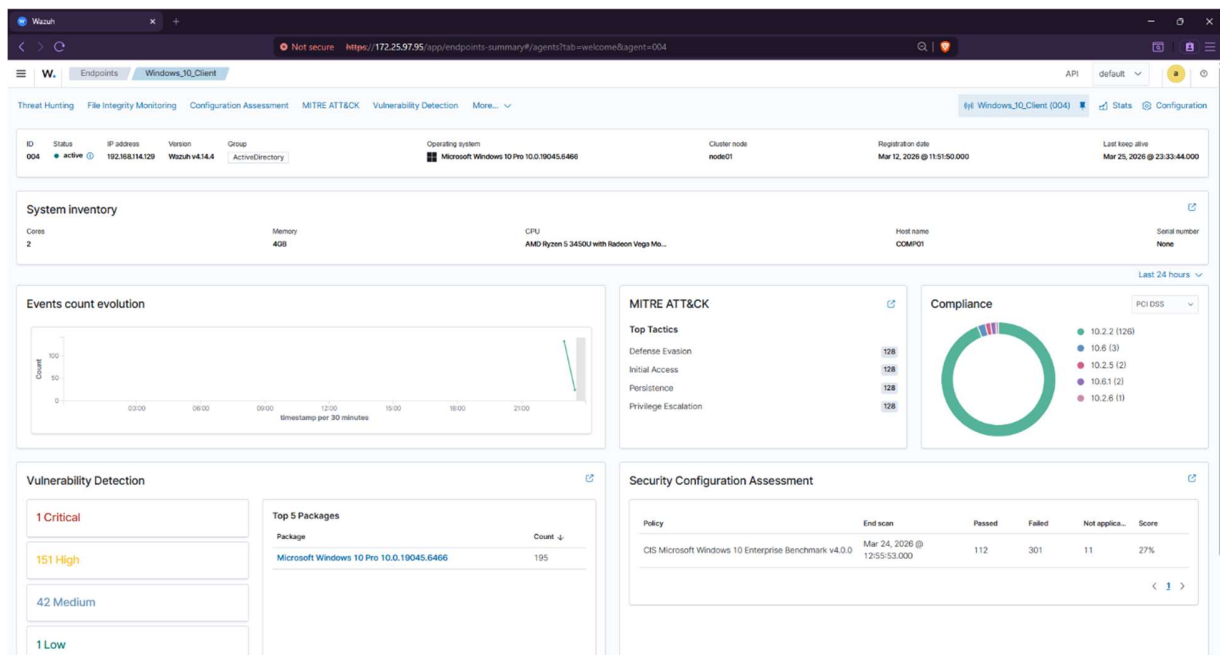
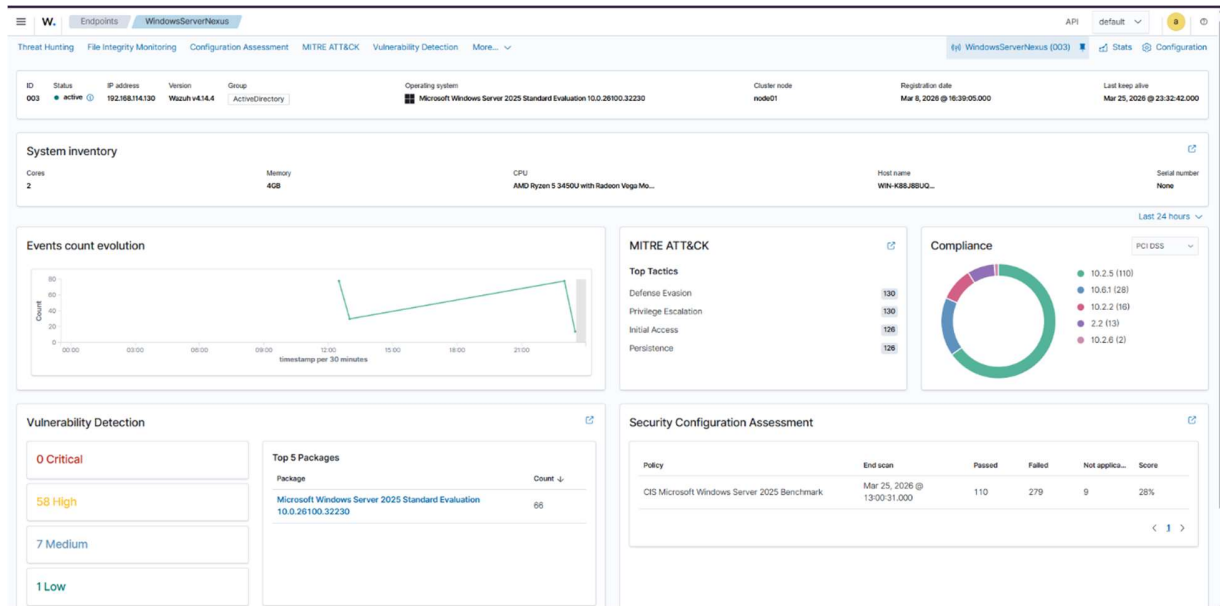
Agents (2)

Deploy new agent Refresh Export formatted More

Search WQL

ID ↑	Name	IP address	Group(s)	Operating system	Cluster node	Version	Status	Actions
003	WindowsServerNexus	192.168.114.130	ActiveDirectory	Microsoft Windows Server 2025 Standard Evaluation 10.0.26100.32230	node01	v4.14.4	active	
004	Windows_10_Client	192.168.114.129	ActiveDirectory	Microsoft Windows 10 Pro 10.0.19045.6466	node01	v4.14.4	active	

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9. Lessons Learned

9.1 Infrastructure & Setup Lessons

- WSL2 networking requires extra attention: By default, WSL2 uses a NAT-based virtual adapter that changes IP on every reboot. A static IP was assigned in the WSL2 configuration to ensure agent-to-manager connectivity remained stable.
- Windows Server 2025 audit policy defaults are sparse: Out of the box, Windows Server 2025 does not log the events needed for effective SIEM monitoring. Advanced Audit Policy via GPO is essential and should be one of the first configurations applied.
- Agent registration can fail silently if the manager is unreachable: Confirming firewall rules allow TCP 1514 from both VMs to the WSL2 IP before attempting agent enrolment saves significant troubleshooting time.
- DNS is foundational: The Windows 10 client was unable to join the domain until its preferred DNS was changed to point to the DC. This is a common pitfall in lab builds and a reminder of how tightly AD depends on DNS.

9.2 Wazuh-Specific Lessons

- Default Wazuh rules are a solid starting point: A significant number of useful alerts fired automatically from the Windows default ruleset (e.g. account creation, privilege use, logon failures) without any custom rule authoring.
- The SCA module is immediately useful: Running the CIS benchmark scan on the first day surfaced two actionable findings (RDP exposed on DC, SMBv1 not disabled) that would otherwise have gone unnoticed.
- Alert noise at low severity is expected in baseline: The bulk of alerts during normal operation are Low/Info severity. Configuring Wazuh Dashboard to filter the default view to Medium and above makes the interface significantly more actionable.
- Wazuh agent groups simplify configuration management: Placing the DC agent in a 'domain-controllers' group and the client in a 'windows-endpoints' group allows group-specific ossec.conf settings (e.g. different FIM paths) to be pushed centrally.

10. Future Improvements

10.1 Short-Term (Next Steps)

- Add a Kali Linux attacker VM to the network and begin executing planned attack simulations
 - Author custom Wazuh detection rules for AD-specific attack patterns such as Kerberoasting (RC4 TGS requests) and DCSync (Event ID 4662 replication calls from non-DC sources)
 - Configure Wazuh email alerts for High and Critical severity events so that significant detections are not missed
 - Expand FIM coverage to include the Windows Registry hives on both machines
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10.2 Long-Term

- Integrate Wazuh with TheHive or a similar incident management platform for structured alert triage and case management
- Evaluate the addition of a second SIEM (e.g. Elastic SIEM or Splunk Free) in parallel to compare detection coverage and operational experience
- Expand the lab to include Azure AD (Entra ID) to simulate a hybrid identity environment and test cloud-side monitoring

Strategic Recommendation

The current lab provides a well-instrumented foundation. The combination of rich telemetry and adversarial testing will rapidly mature the detection capability and produce meaningful, validated security use-cases grounded in real observable data.

Appendix

A. Key Windows Event IDs for AD Monitoring

Event ID	Description
4624	Successful account logon
4625	Failed account logon
4648	Logon attempt using explicit credentials
4662	Operation performed on an AD object (used in DCSync detection)
4672	Special privileges assigned to a new logon (admin logon)
4698	Scheduled task created
4720	User account created
4722	User account enabled
4725	User account disabled
4726	User account deleted
4728/4732	Member added to security / domain global group
4740	User account locked out
4768	Kerberos TGT (AS-REQ) requested
4769	Kerberos TGS ticket requested
4776	NTLM credential validation attempt on DC
5136	Directory service object modified (GPO, OU, AD object changes)
5141	Directory service object deleted
7045	New service installed in the system (used in PsExec detection)

B. Tools & Software Versions

Tool / Software	Version
Wazuh Manager	4.14.x
Wazuh Indexer	4.14.x (OpenSearch)
Wazuh Dashboard	4.14.x

Windows Server 2025	Datacenter Evaluation — Build 26100
Windows 10 Enterprise	22H2
Ubuntu (WSL2)	22.04 LTS
VMware Workstation	Pro (latest)

C. References

- Wazuh Documentation: <https://documentation.wazuh.com>
 - MITRE ATT&CK Framework: <https://attack.mitre.org>
 - Microsoft Advanced Audit Policy: <https://docs.microsoft.com/en-us/windows/security/threat-protection/auditing/advanced-security-auditing>
 - SwiftOnSecurity Sysmon Config: <https://github.com/SwiftOnSecurity/sysmon-config>
 - CIS Benchmarks for Windows Server: https://www.cisecurity.org/benchmark/microsoft_windows_server
 - Wazuh SCA Documentation: <https://documentation.wazuh.com/current/user-manual/capabilities/sec-config-assessment>
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